

The Imaginary Part Compressed into a Coefficient Axis: Where the Product of Imaginary Parts Appears (Exercise XXXIII)

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Abstract

We formalize what happens when the imaginary part is compressed into a coefficient axis. In the complex plane, the imaginary axis does not represent the imaginary part itself; it represents the real coefficients of the imaginary part. The unit i is pressed into the axis's semantics. We prove, with five machine-checked theorems, the consequence of this compression: in multiplication, the product of two imaginary-part coefficients bd leaves the imaginary part and appears in the real part, with a sign determined by the square of the unit. In $\mathbb{C}$ ($i^2 = -1$): the real part of $(a + bi)(c + di)$ is $ac - bd$. In the split-complex frame ($j^2 = +1$): the real part of $(a + bj)(c + dj)$ is $ac + bd$. The cross terms $ad + bc$ stay in the imaginary part in both. The leak sign is the unit square: $i \cdot i = -1$ versus $j \cdot j = +1$. Zero errors, zero sorry.

Keywords: Lean, formalization, complex multiplication, coefficient compression, product leak, 0pat

1 The compression

The imaginary axis of the complex plane displays, at each tick, a real number t — the coefficient of the imaginary part. The imaginary part itself, $t \cdot i$, is the product of this visible coefficient and the invisible unit i . The unit is compressed into the axis: the axis shows coefficients, not units.

This exercise proves what happens under this compression, in multiplication. Two numbers $a + bi$ and $c + di$ multiply to $(ac - bd) + (ad + bc)i$. The observation: the product of the two imaginary-part coefficients, bd , does *not* stay in the imaginary part. It appears in the real part, with a sign. That sign is the square of the unit.

2 The theorems

The axis shows coefficients (complex_imag_axis_coefficient). A point z of the imaginary axis satisfies $z = t \cdot i$ exactly when $z.\text{re} = 0$ and $z.\text{im} = t$. The tick t is a real coefficient; the unit i is not displayed by the axis.

The product of imaginary parts leaks to the real part (complex_mul_imag_leaks). The real part of $(a+bi)(c+di)$ is $ac - bd$. The term bd is the product of the two imaginary-part coefficients; it appears in the real part with a minus sign, because $i \cdot i = -1$.

The cross terms stay in the imaginary part (complex_mul_cross_stays_imag). The imaginary part is $ad + bc$: one imaginary coefficient

times one real coefficient stays in the imaginary part. Only the product of two imaginary coefficients leaks.

The split frame leaks with the opposite sign (split_mul_imag_leak). In the split-complex frame ($j^2 = +1$), the real part of $(a + bj)(c + dj)$ is $ac + bd$. The same compression, the same leak — but the sign is now plus, because $j \cdot j = +1$.

The leak sign is the unit square (leak_sign_is_unit_square). $i \cdot i = -1$ in $\mathbb{C}$ and $j \cdot j = +1$ in the split frame. The sign with which the product of imaginary coefficients appears in the real part is exactly the square of the unit that was compressed into the axis.

3 Honest boundary and position

The content is elementary algebra (multiplication formulas); the value is the precise statement of the compression's consequence. The result concerns the coordinate structure of the frames; no claim is made about the Riemann zeta function. The analysis of ζ remains the open span of the bridge.

4 Formalization

The proof is a single Lean file (proof.lean, 95 lines), importing Mathlib.Data.Complex.Basic and Mathlib.Data.Real.Basic. Five theorems, compiled with Lean 4.32.2 / mathlib v4.32.2, zero errors, zero sorry.

Theorem	Statement		P
complex_imag_axis_coefficient	$z = t \cdot i \leftrightarrow \text{re } z = 0 \wedge \text{im } z = t$	92	C
complex_mul_imag_leaks	$\text{re}((a + bi)(c + di)) = ac - bd$	93	s
complex_mul_cross_stays_imag	$\text{im}((a + bi)(c + di)) = ad + bc$	94	s
split_mul_imag_leaks	$\text{re}((a + bj)(c + dj)) = ac + bd$	95	r
leak_sign_is_unit_square	$i \cdot i = -1, j \cdot j = +1$	96	e

5 Conclusion

The compression of the imaginary part into a coefficient axis has one precise, machine-checked consequence: in multiplication, the product of two imaginary-part coefficients leaks into the real part, and the sign of the leak is the square of the unit compressed into the axis (-1 for $\mathbb{C}$, $+1$ for the split frame). The cross terms stay in the imaginary part. The frames are fully understood; the analysis of ζ remains the open span of the bridge.

References